IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

SELECTED TOPICS IN POWER SYSTEMS

Wednesday, 8 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. Question 1 & 2 should be answered in Booklet A,

Questions 3 & 4 in Booklet B.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Answer ALL Questions

Question 1 on Flexible AC Transmission System (FACTS)

A 400 kV line is transmitting 2.0 p.u. active power. The line reactance is 0.3 p.u. and the voltages at the two ends of the line are maintained at 1.0 p.u. Answer the following questions treating each case in isolation. Assume a self-clearing fault with negligible power flow through the line during the fault and neglect the resistance and shunt susceptance of the line.

a) How much is the critical clearing angle (in degrees) in the base case with no line compensation?

[2]

b) How much is the steady-state stability limit if a STATCOM having a rated current of 4.2 p.u. is used to regulate the voltage at the mid-point of the line at 1.0 p.u.?

Use the formulae provided below:

Reactive power exchanged with the STATCOM

$$Q_C = \frac{4}{X}\left(1 - \cos\frac{\delta}{2}\right)$$

Power-angle relationship once the rated current limit (I_{max}) of the STATCOM is reached

$$P = \frac{V^2}{X}\sin\delta + \frac{VI_{max}}{2}\sin\frac{\delta}{2}$$

[7]

c) How much is the critical clearing angle (in degrees) if an SVC having a rated susceptance of 4.75 p.u. is used to regulate the voltage at the mid-point of the line at 1.0 p.u.?

Use the formulae provided below:

Reactive power exchanged with the SVC

$$Q_C = \frac{4}{X}\left(1 - \cos\frac{\delta}{2}\right)$$

Power-angle relationship once the rated susceptance limit (B_{max}) of the SVC is reached

$$P = \frac{4}{X(4 - XB_{max})}\sin\delta$$

[8]

d) How much is the critical clearing angle (in degrees) with 30% series compensation?

[3]

[20 marks]

Question 2 on System Operation with Low Inertia

a) To maintain frequency stability in the future low-inertia power system, it is important to meet the system frequency requirements after large disturbances. The system dynamic is complicated but can be approximately described by the following swing equation:

$$2H\dot{f}(t) = R(t) - \Delta P_L$$

i) Draw a typical post-disturbance frequency evolution curve and identify the key factors influencing system frequency performance according to the regulation in GB.

[3]

ii) The constraint that guarantees the system frequency nadir within a maximum admissible frequency deviation is given below:

$$\left(H - \mathrm{EFR} \times \frac{\mathrm{T_{EFR}}}{4 \times \Delta f_{\max}}\right) \times R \geq \frac{(P_{\mathrm{L}} - \mathrm{EFR})^2 \times \mathrm{T_d}}{4 \times \Delta f_{\max}}$$

Nomenclature

H: System level of inertia (MWs/Hz)
R: Amount of frequency response scheduled (MW)
EFR: Amount of Enhance Frequency Response (also called Dynamic Containment) scheduled (MW)
P_{L}: Value of generation outage we want to secure against (MW)
$\Delta f_{\max}$: Maximum admissible frequency deviation to avoid load shedding
$\mathrm{T_d}$: Delivery time of frequency response (s)
$\mathrm{T_{EFR}}$: Delivery time of EFR (s)

Other assumptions:

- Numerical values: H = 3 GWs/Hz, $\Delta f_{\max}$ = 0.8 Hz, $\mathrm{T_d}$ = 9.6 s, $\mathrm{T_{EFR}}$ = 0.64 s

1) The system operator has to procure the optimal amount of EFR and R to maintain the nadir constraint. In one operational time slot, the procured EFR and PFR are 0.5 GW and 1.1 GW respectively. Find the generation outage the operator wants to secure against.

[2]

2) In the case of (1), compare the effectiveness of increasing EFR, PFR and reducing $\mathrm{P_L}$ in maintaining the frequency nadir constraint quantitively. Explain the reasons behind the comparison. (Hint: find the partial derivatives among the three quantities.)

[5]

b) Answer the following questions related to the implementation of frequency support services in renewable generation systems:

i) Explain what control mechanisms can a variable-speed wind turbine use to create a headroom (ie be able to ramp up its active power when required) to provide frequency-support services.

[4]

ii) A wind turbine with a power rating of 8 MW operates above rated wind speed with a deloading of 1 MW (ie it generates 7 MW). Calculate the highest synthetic inertia constant, H, the wind turbine could emulate under this operating point if the worst RoCoF expected is −2 Hz/s and the system's nominal frequency is 50 Hz.

[3]

iii) Draw the approximate graph of the kinetic energy stored in a wind turbine versus the wind speed. Explain the shape of the curve.

[3]

[20 marks]

Question 3 on Power System State Estimation

a)

i) What is the purpose of state estimation in power systems?

[4]

ii) Why can't power flow replace the state estimation?

[3]

iii) How do the errors differ from bad data in a state estimation?

[3]

b) In the context of a power distribution system

i) What is real measurement? Provide examples of such measurements.

[2]

ii) What is pseudo measurement? Provide examples of such measurements.

[2]

iii) What is zero injection?

[2]

iv) How are these measurements represented in the state estimation objective function?

[4]

[20 marks]

Question 4 on Electricity Market and Peer-to-Peer Trading

a) In an electricity market, there are 6 generation units. The units labelled "S" belong to a strategic producer, while those labelled "O" belong to other non-strategic producers who are price-takers. For the given hour, assume that wind farm "O1" is operating at 70% of its capacity. Additionally, "D" represents price-elastic demand. The operational details of the generation units and demands are as follows:

Table 4.1: Operational details of generation units

Unit	S1	S2	S3	O1	O2	O3
Technology	Coal	Gas	Gas	Wind	Gas	Gas
Cost (£/MWh)	15.2	23.4	19.1	0	20.1	24.7
Capacity (MW)	170	100	200	400	210	80

Table 4.2: Operational details of demands

Unit	D1	D2	D3
Bid price (£/MWh)	26.5	23.7	22.5
Quantity (MW)	200	500	200

i) Perfect competition: assuming all producers offer their true cost, draw the market demand-supply curve, calculate the generation and demand schedules, market-clearing price, generation profits, demand benefits, and the value of social welfare.

[5]

ii) Strategic offering: assuming the strategic producer owing units S1-S3 offers strategically at 1.2 times their true cost, draw the market demand-supply curve, calculate the generation and demand schedules, market-clearing price, generation profits, demand benefits, and the value of social welfare.

[5]

b) In a local energy market, 3 consumers (deficit) and 3 prosumers (surplus) participate in a double-auction market, where the order book is organised in Table 4.3. The grid buy price and grid sell price are 0.18 £/kWh and 0.04 £/kWh, respectively.

Table 4.3: Order book of the double-auction market

Consumer/Prosumer	C1	C2	C3	P1	P2	P3
Bid/Offer price (£/kWh)	0.17	0.14	0.10	0.06	0.08	0.12
Quantity (kW)	2	4	3	4	3	2

i) Calculate all trading activities and update the order book per trading activity.

[5]

ii) Calculate both the physical and economic benefits of this local energy market.

[5]

[20 marks]